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HIGH-VOLTAGE SYSTEM TEST MEASUREMENT SYSTEMS AND METHODS

Abstract

In a method and apparatus, an electrical load is disposed across an electric power source. First and second electrical components are disposed between first and second terminals, respectively, and the load and power source. The first electrical component is configured so that the first component passes at most a low level electric current to the first terminal upon a short circuit condition. The second electrical component is configured so that the second component passes at most a low level electric current to the second terminal upon a short circuit condition.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of U.S. application Ser. No. 16/909,432, filed Jun. 23, 2020, entitled HIGH-VOLTAGE TEST MEASUREMENT SYSTEMS AND METHODS, which claims the benefit of U.S. provisional patent application No. 62/869,447, filed Jul. 1, 2019, and U.S. provisional patent application No. 62/870,010, filed Jul. 2, 2019, entitled HIGH-VOLTAGE TEST MEASUREMENT SYSTEMS AND METHODS, the entire disclosures of each which are hereby incorporated by reference herein for all purposes.

TECHNICAL FIELD

[0002] Embodiments of the present disclosure generally relate to the field of high-voltage electrical systems. More particularly, certain embodiments of the present disclosure relate to improved high-voltage electrical systems wherein a test port is provided so that an operator can ensure that an electrically safe working condition exists prior to performing maintenance on the high-voltage electrical system. Certain embodiments of the present disclosure may find use in electrically-powered vehicles, among other applications.

BACKGROUND

[0003] High-voltage electrical systems are found in a variety of applications, including in electrically-powered vehicles. In general, a high-voltage electrical system may include a power source, such as mains power or one or more batteries, that serve an electrical load. As an example, a battery powered electric bus might have a plurality of high-voltage battery packs that serve a variety of high-voltage components or systems on the bus, such as an electric drive train, a heating, ventilation, and air conditioning (HVAC) system, an air compressor, a coolant heater, and DC/AC inverters. These high-voltage components may operate at voltages up to seven hundred and fifty volts direct current (750 VDC). Other high-voltage electrical systems have other components that operate at similarly high, or higher, voltages.

[0004] As those of skill in the art will appreciate, the electric current that flows in a high-voltage electrical system may also be high, and human operators must exercise caution when performing maintenance on the high-voltage electrical system components and/or the electric power source. Even if the overall system is shut down or otherwise non-operational, however, dangerously high voltages can still be present in the high-voltage electrical system because the electric power source is not necessarily disconnected from all or part of the electrical circuitry. Also, dangerously high voltages can still be present because of secondary sources of residual energy, such as flywheels, capacitors, inductors, and motor back EMF, among others.

[0005] Accordingly, a typical high-voltage electrical system includes one or more safety systems, such as switches (including electrical and manual disconnects, electrical contactors, and interlocks) at various points in the electrical circuitry. Prior to working on the high-voltage electrical system, an operator may perform a “lockout/tagout” (LOTO) procedure, including toggling one or more manual switches and locking them in the disconnected position, to prevent unexpected energization of the system while maintenance is being performed. While this procedure is intended to create an electrically safe working condition, after following it the operator must nonetheless verify that the high-voltage electrical system is electrically safe prior to beginning work. The operator may use a test instrument (e.g., a multimeter) to take measurements of electrical characteristics at various

points in the high-voltage electrical system to ensure hazardous voltages and currents are not still present in the system. Due to the risk of high current present in the system, the operator may be required to wear various safety gear, including a full body arc-flash suit. The suit is to protect the operator from electrocution caused by electric arcs between the electrical system and the operator. [0006] The foregoing discussion is intended only to illustrate various aspects of the related art in the field of the disclosure at the time, and should not be taken as a disavowal of claim scope.

SUMMARY

[0007] According to an embodiment, a high-voltage electrical system has a high-voltage electric power source, an electrical load disposed electrically across the electric power source so that the electric power source provides electric power to the electrical load, a first electrical terminal, at least one first electrical component, and a second electrical terminal and at least one second electrical component. The first at least one electrical component is disposed and configured so that the first at least one electrical component passes at most a low level electric current to the first electrical terminal when a short circuit between the first electrical terminal and a portion of an electric circuit including the electric power source and the electrical load that has a high voltage level completes an electric circuit that includes the first at least one electrical component and the electric power source. The second at least one electrical component is disposed and configured so that the second at least one electrical component passes at most a low level electric current toward the second electrical terminal when a short circuit between a portion of an electric circuit including the electric power source and the electrical load that has a high voltage level completes an electric circuit that includes the second at least one electrical component and the electric power source.

[0008] In another embodiment, a method of testing a high-voltage electrical system within an electric bus includes providing a body supported by a plurality of wheels, at least one electric motor disposed so that the at least one electric motor drives the plurality of wheels, a high-voltage electric power source in electrical communication with the at least one electric motor, an electrical load, including the at least one electric motor, disposed electrically across the electric power source so that the electric power source provides electric power to the electrical load, a first electrical terminal and at least one first electrical component, and a second electrical terminal and at least one second electrical component. The first at least one electrical component is disposed and configured so that the first at least one electrical component passes at most a low level electric current toward the first electrical terminal when a short circuit between the first electrical terminal and a portion of an electric circuit including the electric power source and the electrical load that has a high voltage level completes an electric circuit that includes the first at least one electrical component and the electric power source. The second at least one electrical component is disposed and configured so that the second at least one electrical component passes at most a low level electric current toward the second electrical terminal when a short circuit between a portion of an electric circuit including the electric power source and the electrical load that has a high voltage level completes an electric circuit that includes the second at least one electrical component and the electric power source. A voltage test meter is operatively connected electrically across the first electrical terminal and the second electrical terminal.

[0009] In a further embodiment, a high-voltage electrical system has a high-voltage electric power source, an electrical load disposed electrically across the electric power source so that the electric power source provides electric power to the electrical load, a first electrical terminal, a second electrical terminal, a first at least one electrical component electrically disposed between the first electrical terminal and the electric power source and between the first electrical terminal and the electrical load, and a second at least one electrical component electrically disposed between the second electrical terminal and the electric power source and between the second electrical terminal and the electrical load. The first at least one electrical component is configured so that the first at least one electrical component passes at most a low level electric current to the first electrical terminal when a short circuit between the first electrical terminal and a portion of an electric circuit

including the electric power source and the electrical load that has a high voltage level completes an electric circuit that includes the first at least one electrical component and the electric power source. The second at least one electrical component is configured so that the second at least one electrical component passes at most a low level electric current to the second electrical terminal when a short circuit between a portion of an electric circuit including the electric power source and the electrical load that has a high voltage level completes an electric circuit that includes the second at least one electrical component and the electric power source.

[0010] In another embodiment, an electric bus has a body supported by a plurality of wheels, at least one electric motor disposed so that the at least one electric motor drives the plurality of wheels, a high-voltage electric power source in electrical communication with the at least one electric motor, an electrical load, including the at least one electric motor, disposed electrically across the electric power source so that the electric power source provides electric power to the electrical load, a first electrical terminal, a second electrical terminal, a first at least one electrical component electrically disposed between the first electrical terminal and the electric power source and between the first electrical terminal and the electrical load, and a second at least one electrical component electrically disposed between the second electrical terminal and the electric power source and between the second electrical terminal and the electrical load. The first at least one electrical component is configured so that the first at least one electrical component passes at most a low level electric current to the first electrical terminal when a short circuit between the first electrical terminal and a portion of an electric circuit including the electric power source and the electrical load that has a high voltage level completes an electric circuit that includes the first at least one electrical component and the electric power source. The second at least one electrical component is configured so that the second at least one electrical component passes at most a low level electric current to the second electrical terminal when a short circuit between a portion of an electric circuit including the electric power source and the electrical load that has a high voltage level completes an electric circuit that includes the second at least one electrical component and the electric power source.

[0011] In a further embodiment, a high-voltage electrical system has a high voltage electric power source, and an electrical load. Electrical circuitry includes the electric power source and the electrical load so that the electric power source applies high voltage power to the electrical load. At least one switch is disposed in the electrical circuitry upstream from the electrical load with respect to the electric power source so that the at least one switch, in a first switch state, conducts electric current within the electrical circuitry through the at least one switch and, in a second switch state, opens at least part of the electrical circuitry so that the high voltage power is not applied to at least part of the electrical load. A first electrical terminal is connected to a first position, between the at least one switch and the electrical load, in a portion of the electrical circuitry at which high voltage exists when the at least one switch is in the first switch state. A second electrical terminal is connected to a second position, between the at least one switch and the electrical load, in the portion of the electrical circuitry at which high voltage exists when the at least one switch is in the first switch state. Application of a predetermined resistance across the first and second electrical terminals produces a voltage drop across the predetermined resistance that is in a voltage drop first state when the at least one switch is in its first switch state and is in a voltage drop second state when the at least one switch is in its second switch state. The voltage drop first state is different from the voltage drop second state. A first at least one electrical component is electrically disposed between the first electrical terminal and the first position in the electrical circuitry so that the first at least one electrical component passes at most a low level of electric current from the first position to the first electrical terminal when a short circuit between the first electrical terminal and the portion of the electrical circuitry completes an electric circuit that includes the first at least one electrical component and the electric power source. A second at least one electrical component is electrically disposed between the second electrical terminal and the second position in the electrical

circuitry so that the second at least one electrical component passes at most a low level of electric current from the second position to the second electrical terminal when a short circuit between the second electrical terminal and the portion of the electrical circuitry completes an electric circuit that includes the second at least one electrical component and the electric power source.

[0012] In a still further embodiment, an electric bus has a body supported by a plurality of wheels, at least one electric motor disposed so that the at least one electric motor drives the plurality of wheels, and a high voltage electric power source supported by the body and in electrical communication with the at least one electric motor, wherein the electric power source comprises one or more battery packs. An electrical load includes the at least one electric motor. Electrical circuitry includes the electric power source and the electrical load so that the electric power source applies high voltage power to the electrical load. At least one switch is disposed in the electrical circuitry upstream from the electrical load with respect to the electric power source so that the at least one switch, in a first switch state, conducts electric current within the electrical circuitry through the at least one switch and, in a second switch state, opens at least part of the electrical circuitry so that the high voltage power is not applied to at least part of the electrical load. A first electrical terminal is connected to a first position, between the at least one switch and the electrical load, in a portion of the electrical circuitry at which high voltage exists when the at least one switch is in the first switch state. A second electrical terminal is connected to a second position, between the at least one switch and the electrical load, in the portion of the electrical circuitry at which high voltage exists when the at least one switch is in the first switch state. Application of a predetermined resistance across the first and second electrical terminals produces a voltage drop across the predetermined resistance that is in a voltage drop first state when the at least one switch is in its first switch state and is in a voltage drop second state when the at least one switch is in its second switch state. The voltage drop first state is different from the voltage drop second state. A first at least one electrical component is electrically disposed between the first electrical terminal and the first position in the electrical circuitry so that the first at least one electrical component passes at most a low level of electric current from the first position of the electrical circuitry to the first electrical terminal when a short circuit between the first electrical terminal and the portion of the electrical circuitry completes an electric circuit that includes the first at least one electrical component and the electric power source. A second at least one electrical component is electrically disposed between the second electrical terminal and the second position in the electrical circuitry so that the second at least one electrical component passes at most a low level electric current from the second position to the second electrical terminal when a short circuit between the second electrical terminal and the portion of the electrical circuitry completes an electric circuit that includes the second at least one electrical component and the electric power source.

[0013] In another embodiment, a high-voltage electrical system has a high-voltage electric power source, an electrical load disposed electrically across the electric power source so that the electric power source provides electric power to the electrical load, and a test port and at least one electrical component. The at least one electrical component is disposed and configured so that the at least one electrical component passes at most a low level electric current to the test port when a short circuit between the test port and a portion of an electric circuit including the electric power source and the electrical load that has a high voltage level completes an electric circuit that includes the at least one electrical component and the electric power source.

[0014] In a further embodiment, an electric bus has a body supported by a plurality of wheels, at least one electric motor disposed so that the at least one electric motor drives the plurality of wheels, a high-voltage electric power source in electrical communication with the at least one electric motor, an electrical load, including the at least one electric motor, disposed electrically across the electric power source so that the electric power source provides electric power to the electrical load, and a test port and at least one electrical component. The at least one electrical component is disposed and configured so that the at least one electrical component passes at most a

low level electric current to the test port when a short circuit between the test port and a portion of an electric circuit including the electric power source and the electrical load that has a high voltage level completes an electric circuit that includes the at least one electrical component and the electric power source.

[0015] An embodiment of a method of testing a high-voltage electrical system within an electric bus includes the step of providing a body supported by a plurality of wheels, at least one electric motor disposed so that the at least one electric motor drives the plurality of wheels, a high-voltage electric power source in electrical communication with the at least one electric motor, an electrical load, including the at least one electric motor, disposed electrically across the electric power source so that the electric power source provides electric power to the electrical load, and a test port and at least one electrical component. The at least one electrical component is disposed and configured so that the at least one electrical component passes at most a low level electric current to the test port when a short circuit between the test port and a portion of an electric circuit including the electric power source and the electrical load that has a high voltage level completes an electric circuit that includes the at least one electrical component and the electric power source. The method includes operatively connecting a voltage test meter electrically to the test port.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] Having thus described some example embodiments in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0017] FIG. 1 is a schematic illustration of circuit diagram of a high-voltage electrical system in accordance with an embodiment of the present disclosure;

[0018] FIG. 2 is a schematic illustration of a circuit diagram of a high-voltage electrical system in accordance with an embodiment of the present disclosure;

[0019] FIG. 3 is a perspective view of one example of an electrically-powered vehicle with which embodiments of the present disclosure may be used;

[0020] FIGS. 4A-4C present a block diagram of one example of a high-voltage and accessory architecture of an electrically-powered vehicle with which embodiments of the present disclosure may be used;

[0021] FIGS. 5A-5C present a circuit diagram of one example of a high-voltage interlock loop that may be used with the high-voltage and accessory architecture of FIGS. 4A-4C;

[0022] FIG. 6 is a perspective view of one example of a low voltage junction box including one example of a low voltage disconnect that may be used with the high-voltage and accessory architecture of FIGS. 4A-4C;

[0023] FIGS. 7 and 8 are perspective views of opposing sides of one example of an energy storage system junction box comprising a test port in accordance with an embodiment of the present disclosure;

[0024] FIG. 9 is a detail view of one example of an energy storage system junction box comprising a test port in accordance with an embodiment of the present disclosure, wherein test port access covers are engaged; and

[0025] FIG. 10 is a perspective view of the energy storage system junction box of FIG. 9 wherein an operator is using the test port with a multimeter.

[0026] Repeat use of reference characters in the present specification and drawings is intended to represent same or analogous features or elements of embodiments of the present invention.

DETAILED DESCRIPTION

[0027] Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. Each example is provided by way

of explanation of the disclosure, not limitation of the disclosure. In fact, it will be apparent to those skilled in the art that modifications and variations can be made in the present disclosure without departing from the scope or spirit thereof. For instance, any number of features illustrated or described as part of one embodiment may be used on another embodiment, in any combination, to yield a still further embodiment. Thus, it is intended that the present disclosure covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0028] Further, either of the terms “or” and “one of _____ and _____,” as used in this disclosure and the appended claims is intended to mean an inclusive “or” rather than an exclusive “or.” That is, unless specified otherwise, or clear from the context, either of the phrases “X employs A or B” and “X employs one of A and B” is intended to mean any of the natural inclusive permutations. That is, either phrase is satisfied by any of the following instances: X employs A; X employs B; or X employs both A and B, regardless whether the phrases “at least one of A or B” or “at least one of A and B” are otherwise utilized in the specification or claims. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from the context to be directed to a singular form. Throughout the specification and claims, the following terms take at least the meanings explicitly associated herein, unless the context dictates otherwise. The meanings identified below do not necessarily limit the terms, but merely provide illustrative examples for the terms. The meaning of “a,” “an,” and “the” may include plural references, and the meaning of “in” may include “in” and “on.” The phrase “in one embodiment,” or the like, as used herein does not necessarily refer to the same embodiment, although it may.

[0029] Embodiments of the present disclosure comprise apparatus, systems, and methods for performing test measurements on and/or ensuring that an electrically safe working condition exists in a high-voltage electrical system. In various embodiments, a test port is provided as part of the electrical circuitry of a high-voltage electrical system so that an operator can ensure that high voltage is not present in some or all of the high-voltage electrical system. The high-voltage electrical system comprises a high-voltage electric power source that may have a ground connection or that may be electrically isolated from non-circuitry components of the environment in which the high-voltage system exists. The high-voltage source applies power to an electrical load, with the result that there is at least part of the overall circuitry within the system that that, when operative, may have high voltages and may carry a high electric current between the high voltage power source and the load. In some embodiments, the test port is accessible to a human operator. In such embodiments, the test port is disposed in electrical communication with that portion of the electrical circuitry at which high voltages exist so that, in the event of a short circuit (e.g. caused by or through the human operator) between the test port and the power delivery portion of the circuitry that completes an electric circuit that includes the highest level power source in the circuitry, the electric current at the test port is low. In the embodiments discussed herein, the test port is electrically connected to a portion of the power delivery circuitry at which the highest possible voltage levels could occur. Thus, for example, where the highest possible voltage level in the power delivery circuitry is known, and an assumption is made that this highest possible voltage level will be present at the point at which the test port connects to the power delivery circuitry and that the lowest voltage point will be at the terminal, then a resistance may be disposed electrically in series between the test port and that connection so that the electric current at the test port is, at most, a low level electric current so that the human operator should not be injured. In the embodiments discussed below, the highest voltage level in the power delivery circuitry is approximately 750V, and the resistance between that circuitry and the test port is approximately 150 kΩ or greater, so that under Ohm's Law, the electric current at the test port is, at most, 5 mA. In certain embodiments, the high-voltage electrical system is a component of an electrically-powered vehicle, including but not limited to a passenger bus, and the electric power source includes one or

more battery packs. In some embodiments, use of the test ports may allow an operator to test the high-voltage electrical system without wearing an arc-flash suit.

[0030] Although some embodiments are discussed below in the context of a battery electric bus, those of skill in the art will appreciate that the present disclosure is not so limited. In particular, it is contemplated that embodiments of the present disclosure may be used with any suitable high-voltage electrical system, including high-voltage electrical systems in vehicles, buildings, power plants and grids, among other applications. Electrically-powered vehicles with which embodiments of the present disclosure may be used include, but are not limited to, battery electric and/or gasoline- or diesel-electric hybrid cars, trucks, and buses, or any other vehicle having electrical systems where high voltages are present.

[0031] In this regard, FIGS. 1-2 present schematic circuit diagrams of high-voltage electrical systems in accordance with various embodiments of the present disclosure. Turning first to FIG. 1, a high-voltage electrical system **40** includes an electric power source **12** and an electrical load **14**. In various embodiments, electric power source **12** is a high-voltage power source operative to supply power at high voltages to electrical load **14**. In the embodiment shown in FIG. 1, power source **12** is a seven hundred and fifty volts direct current (750 VDC) power source, e.g. used in an electric vehicle such as a bus, that is electrically isolated from vehicle systems other than those incorporated in the vehicle's high-voltage system. Thus, e.g., power source **12** is electrically isolated from the vehicle chassis and is thereby isolated from chassis ground.

[0032] As used herein, the terms “high-voltage” and “high-voltage power” refer to electrical energy at voltage drops across individual components or groups of components of the electrical load above approximately fifty volts AC (50 VAC) or fifty volts DC (50 VDC). As used herein, an “electric power source” may include, e.g., a battery source such as one or more battery packs, mains power, an AC power source, and related circuitry, e.g. inverter or resistor networks, transformers, or power supplies, to provide an operative power supply to an electrical load. In various embodiments, an electric power source may also include secondary sources of residual energy, such as flywheels, capacitors, inductors (or components with capacitors or inductors), and/or motor back EMF. An “electrical load” comprises one or more electrical components of a high-voltage electrical system that consume electric power from the electric power source in the system's normal operation. In the illustrated example, high-voltage electrical system **40** includes electrical circuitry **16** that provides electrical communication between, and includes, electric power source **12** and electrical load **14**. Thereby, electric power source **12** applies high-voltage power to electrical load **14**.

[0033] Also as shown in FIG. 1, a switch **18** is disposed in electrical circuitry **16** upstream from electrical load **14** with respect to electric power source **12**. As used herein, the term “switch,” with regard to electrical circuits, refers to a device that operatively exists in at least two states, one of which opens flow of electric current in at least a part of the electrical circuitry, and one of which conducts such flow. The switch, e.g. a lever-operated switch or an electrical contactor, may be self-contained in a housing, such that the actuator, e.g. mechanical force applied by an operator to a switch handle or electromagnetic force, changes the switch's state without removal of components from the housing. A switch, e.g. a manual service disconnect (MSD), may also be operated by removing a part of the switch to open the switch and replacing the part to close the switch. Thus, it will be understood that the switch may comprise various current disconnect devices operated, e.g., by any of various electrical, mechanical, magnetic, electromechanical, or electromagnetic components and arrangements. Accordingly, in high-voltage electrical system **40** switch **18**, in a first (e.g., closed) state, conducts electric current within electrical circuitry **16** through switch **18**. In a second (e.g., open) state, switch **18** opens at least part of electrical circuitry **16** so that the high voltage power is not applied to at least part of electrical load **14**. Although a single switch **18** is shown in FIG. 1, those of skill in the art will appreciate that high-voltage electrical system **40** comprises at least one switch **18** and, in some embodiments, will comprise a plurality of switches **18**. As described in greater detail below, switch(es) **18** may include, among other things, safety

devices an operator may use prior to performing maintenance on system **40** or components thereof. [0034] As noted above, in embodiments of the present disclosure, a high-voltage electrical system may comprise at least one test port to allow an operator to determine whether an electrically safe working condition exists in the electrical system, or a portion thereof. The test port comprises a pair of electrical terminals. The terminals are respectively connected to the power delivery portion of electrical circuitry **16** at positions opposing each other across at least one component of the electrical load between the at least one switch and the electrical load. For instance, as shown in FIG. **1**, high-voltage electrical system **40** has a test port comprising two terminals **20**, **22** that connect into the high-current-carrying portion of electrical circuitry **16** at positions **24** and **26**, respectively. Terminals **20**, **22** may be any suitable point connections for closing an electrical circuit. Positions **24**, **26** comprise any suitable electrical connection known to those of skill in the art. It should also be understood that, in the embodiment illustrated in FIG. **1** (and similarly to FIG. **3**), electric current will always flow through the circuit branch defined by resistors **42**, **44**, and **46**. In other embodiments, a switch, e.g. a high voltage relay, may be provided in this circuit branch in series, e.g. adjacent connection point **24**, that will be normally open but that the operator will close in order to conduct a test as set forth herein.

[0035] Additionally, and as described in greater detail below, in the embodiment shown in FIG. **1**, electrical circuitry **16** of high-voltage electrical system **40** also comprises resistors **42**, **44**, and **46**. Resistors **42**, **46** are electrically disposed in series between a respective terminal **20**, **22** and respective positions **24**, **26**. In this embodiment, the resistance of each resistor **42**, **46** is up to approximately 1 M Ω , while the resistance of resistor **44** is of at least an order of magnitude smaller, e.g. approximately 10.7 k Ω or 150 k Ω , although it should be understood that resistor **44** may have a resistance comparable to or larger than resistances **42**, **46**. As explained in more detail herein, those of skill in the art will appreciate that the resistance values for resistors **42**, **46** may be different in other embodiments, e.g. depending on the voltage of power source **12** and the specific configuration of electrical circuitry **16**, among other factors.

[0036] The test port may take various forms within the scope of the present disclosure, as those of skill in the art will appreciate. In some embodiments, the test port may include terminals disposed within respective suitable connectors (e.g., an IP rated connector that is configured to receive test leads of a test instrument, such as a multimeter) coupled with a junction box that is accessible to an operator. Likewise, and for example, the positions **24**, **26** at which terminals **20**, **22** are connected may be respective busbars disposed within the junction box. The junction box, in turn, allows the operator to interact with the terminals **20**, **22** but prevents the operator's interaction with the circuit components within the junction box. Accordingly, the present discussion addresses the possibility of short circuit conditions at the terminals.

[0037] In one aspect of the present disclosure, the electrical terminals are connected at respective positions across any of various components present in the electrical circuitry through which power source **12** provides electric power to load **14** and that may define high voltages and carry high current. These respective positions are between one or more of the one or more switches and the electrical load such that application of a predetermined resistance across the electrical terminals produces a first voltage drop across the predetermined resistance during the circuit's normal operation when the at least one switch is in its first (e.g., closed) state and produces a second, different voltage drop when the at least one switch is in its second (e.g., open) state. For example, when the at least one switch is closed, the voltage drop across the predetermined resistance may be higher than the voltage drop across the predetermined resistance when the at least one switch is open (in the embodiment of FIG. **1**, the voltage drop should be zero when the switch is open, unless the load includes a power source). As those of skill in the art will appreciate, the component across which the electrical terminals are connected, and the positions within the electrical circuitry at which the terminals are connected, may vary in various embodiments. Many different configurations are within the scope of the present disclosure, provided that the two states of the

voltage drop across such components are distinguishable based on the testing method used by the operator.

[0038] In certain embodiments described herein, e.g., the terminals are attached to positions in the electrical circuitry providing power from between the power source to the load so that the test port terminal circuitry is electrically in parallel with both the electric power source and the electrical load. As such, the voltage drop across a test resistance, e.g. the resistance of resistor **44** in FIG. **1**, disposed electrically across the terminals may be directly related to the open circuit voltage of electric power source **12** (if all of the one or more switches upstream from the measurement terminals is/are closed). Thus, if the values and/or relationship among the test resistance and the resistances/impedances of components between the test resistance and the power delivery circuitry are/is known, and the electric power source's open circuit voltage is known, the relationship between the electric power source's open circuit voltage and the voltage drop measurement produced by the testing device connected across the terminals may be defined, and therefore known, through solving for the testing device resistance's voltage in the resulting voltage divider network. With this information, the operator may define a voltage measurement value at the testing device that may be interpreted as a dividing line between a determination that the electric power source may be electrically connected to the load via the high voltage circuitry and a determination that the electric power source is not so connected. It should be understood, in view of the present disclosure, that the predetermined resistance of the test resistance need not be known by the operator, provided the operator has been informed of the dividing-line voltage.

[0039] In the example of FIG. **1**, connection of terminals **20**, **22** to the power delivery circuitry at positions **24**, **26** places terminals **20**, **22** and the test circuit in parallel with power source **12** and electrical load **14**. As a result, the voltage drop across the resistance of a test instrument attached to and extending across terminals **20**, **22** may be directly related to the open circuit voltage of electric power source **12**, assuming that switch **18** is in its first (e.g., closed) state and taking into consideration the resistor network as illustrated.

[0040] As noted above, it will be understood that other arrangements of the electrical terminal **20**, **22** positions **24**, **26** are possible within the scope of the present disclosure. For example, the electrical terminal positions may be disposed across a resistor or resistor network that is elsewhere in the power delivery circuitry between the one or more switch(es) and the electrical load and that across which the voltage drop, or voltage drop range, is known when all switch(es) are closed and when at least one of the one or more switch(es) is open, such that measurement of the voltage drop across the electrical terminals by a multimeter or other testing device indicates to the operator whether the testing device voltage drop corresponds to the high-voltage circuitry's closed or open states. In any such arrangement, it will be understood in view of the present disclosure that the values or approximate relationships among the resistances/impedances relevant to determining the testing device resistance's voltage drop in view of the electric power source open circuit voltage should be known with respect to the high-voltage electric circuitry's open and closed states, so that the operator may select the measurement device voltage drop demarcation line between the open and closed states.

[0041] Additionally, in another aspect of the present disclosure, the test port is designed so that, in the normal operation of the test circuit (e.g. when a multimeter is applied across the terminals) or in the event of a short circuit between one or both of the terminals and any position in the high voltage, or power delivery, circuitry that completes a circuit that includes a high-voltage power source, the electric current flow at the given terminal will be below a level dangerous to humans. The terms "short" and "short circuit" include electric arcs between high-voltage or high-current electrical circuitry and human operators. Thus, in various embodiments, at least one respective electrical component is electrically disposed between a first electric terminal of the pair of the electrical terminals and the positions at which that terminal is connected to the electrical circuitry through the high voltage power source provides electric power to the load and at which high

voltages may therefore exist (the “power delivery circuitry”) so that the at least one electrical component passes at most a low level electric current from such a connection position to the first electrical terminal when a short circuit exists between the first electrical terminal and a point in the power delivery circuitry that completes an electrical circuit that includes a high voltage power source. Similarly, at least one respective electrical component is electrically disposed between a second electric terminal of the pair of the electrical terminals and the positions at which that terminal is connected to the power delivery circuitry so that the at least one electrical component passes at most a low level electric current from such a connection position to the second electrical terminal when a short circuit exists between the second electrical terminal and a point in the power delivery circuitry that completes an electrical circuit that includes a high voltage power source. As used herein, “low” electric current is approximately five milliamperes (5 mA) or below, while “high” electric current is above approximately 5 mA and particularly above approximately 10 mA. [0042] As shown in FIG. 1, for example, resistor **42** is electrically disposed between electrical terminal **20** (see position **48**) and position **24** in the power delivery circuit. If a short occurs between point **48** and a position in the power delivery circuitry in communication with the negative pole of source **12** (e.g. via the operator) to thereby complete an electric circuit that includes resistor **42** and power source **12**, and assuming the resistance of resistor **42** is approximately 1 M Ω , current through resistor **42** and at terminal **20** would be at most approximately 0.75 mA under Ohm's Law. If the resistance of resistor **42** is approximately 150 k Ω , the current would be at most approximately 5 mA. Similarly, resistors **44** and **46** are electrically disposed between electrical terminal **20** and position **26** in the power delivery circuitry. If a short occurs between point **48** and a position in the power delivery circuitry in communication with the positive pole of source **12** to thereby complete an electric circuit that includes resistors **44**, **46** and power source **12**, and assuming the resistance of resistor **46** is approximately 1 M Ω and that the resistance of resistor **44** is approximately 10.7 k Ω , electric current through resistors **44**, **46** would be at most approximately 0.74 mA. If the resistance of resistor **46** is approximately 150 k Ω , the current would be at most approximately 4.7 mA. With regard to terminal **22**, resistors **42** and **44** are disposed between it (at point **50**) and connection position **24** in the event of a short from point **50** to a position in the power delivery circuitry in communication with the negative pole of source **12** to thereby complete an electric circuit that includes resistors **42** and **44** and power source **12**, while resistor **46** is disposed between it (at point **50**) and connection position **26** in the event of a short from point **50** to a position in the power delivery circuitry in communication with the positive pole of source **12** to thereby complete an electric circuit that includes resistor **46** and power source **12**, with similar, but mirror-image, resulting electric current flows as would occur at the short circuits at point **48** discussed above. Thus, in this embodiment, each of resistors **42** and **46** has a resistance above approximately 150 k Ω and may have a resistance between approximately 150 k Ω and 1 M Ω . The foregoing discussion assumes that the highest possible voltage level in the high current level circuitry that includes the power source and the load is the voltage drop across the power source, in this instance 750 VDC. It should be understood that such discussion is for purposes of explanation only and that if there are additional power sources within the high current portion of the circuitry, the power sources could accumulate in order to define the highest possible voltage level that either of the terminals could see at the point where the terminal communicates with that circuitry at points **24** and **26** (which, in the embodiments discussed herein, is the highest possible voltage that exists in the circuitry as it normally operates). In that event, the value of the intervening series resistor **42**, **46** could be determined under Ohm's Law so that a maximum electric current level at the respective terminal is no more than a maximum low electric current level, e.g. 5 mA.

[0043] The foregoing discussion assumes that switch **18** is in the first (e.g., closed) position. However, if switch **18** is in the second (e.g., open) position, then no current should flow in the event of a short, as described above, or connection of a testing device, such as a multimeter. Assuming an approximately 750 VDC power source **12**, that switch **18** is closed, that the resistance

of each of resistors **42** and **46** is approximately 1 M Ω , and that the resistance of resistor **44** is approximately 10.7 k Ω , if an operator applies the probes of a multimeter to terminals **20** and **22** so that the multimeter measures the voltage across the terminals, the multimeter should detect an approximately 4V voltage drop. If the resistance of resistor **44** is approximately 150 k Ω , the multimeter should detect an approximately 52V voltage drop. If switch **18** is open, the multimeter should detect 0V, so that the distinction between 0V and 4V (or between 0V and 52V, depending on the resistance of test resistor **44**) informs the operator of the operative state of the power delivery circuit. In any event, even if either of terminals **20** and **22** is shorted to a point in the power delivery circuit, the operator applying the multimeter to the test port comprising the terminals should be subject to at most a low level electric current. As should be understood, the inherent resistance of multimeters is typically high (e.g. approximately 15 M Ω), and the multimeter's disposition electrically across terminals **20** and **22** would not materially alter the current determinations set forth in the discussion above.

[0044] Similarly, in some embodiments, an LED voltage tester may be configured with a resistance in such circumstances (where the relationship between that resistance and the in-series resistances is known) that the tester may light the LED at a voltage drop across the testing device resistance that is above approximately four volts (4V) or other level, e.g. 52V, depending on the test resistance value. If, however, the testing device's internal resistance is much lower than the in-series resistances of resistors **28** and **30**, such that the voltage drop read by the testing device will be much lower than the open circuit voltage of electric power source **12** when all switch(es) **18** are closed, the operator may choose a lower voltage drop dividing line to demark the circuitry's closed state from its open state.

[0045] While the discussion above illustrates the use of a DC power source, it should be understood that power source **12** could be an AC power source. It will also be understood, in view of the present disclosure, that the discussion above is applicable to a system having an AC power source at **12**, with regard to short circuits between either of points **48** and **50** and the power source.

[0046] FIG. **2** is an example of a high-voltage electrical system configuration within the scope of the present disclosure. FIG. **2** is a simplified circuit diagram of a high-voltage electrical system **60** that is similar in certain respects to system **40**, described above with respect to FIG. **1**. Thus, in FIGS. **1-2**, like parts are indicated by like reference numerals.

[0047] Referring to FIG. **2**, in this embodiment, high-voltage circuitry **16** of system **60** comprises a transformer **62** to step down the voltage at each electrical terminal relative to the voltage at electric power source **12**, which is an alternating current (AC) power source in this example. In particular, transformer **62**, which comprises a primary winding **64** and a secondary winding **66**, is electrically disposed in series between resistors **68** and **70**, respectively. (Those of skill in the art will appreciate that resistors **68**, **70** need not be provided in all embodiments employing transformer **62**.) Resistors **68** and **70**, which may be analogous to resistors **28** and **30** described above, are connected to high-voltage circuitry **16** at positions **24** and **26**, respectively of FIG. **2**. Electrical terminals **20** and **22** are in electrical communication with secondary winding **66** of transformer **62** in any suitable manner familiar to those of skill in the art. As shown, transformer **62**, resistors **68**, **70**, and terminals **20**, **22** are connected electrically in parallel to power source **12** and load **14**.

[0048] As those of skill in the art will appreciate, and as with the embodiment described above with respect to FIG. **1**, in the embodiment shown in FIG. **2**, terminals **20**, **22** are attached to positions in the electrical circuitry such that the voltage drop across the resistance of a testing device attached to and extending across the terminals will be directly related to the open circuit voltage of the electric power source (if all of the one or more switches upstream from the measurement terminals is/are closed), subject to the impact of any electrical components in series between the measurement device resistance and the electric power source terminals. Likewise, in this embodiment, the test port is designed so that, in the event of a short circuit between one of the terminals **20**, **22** and a position in the power delivery circuit that completes an electric circuit including the transformer

and power source **12**, any current flowing through the completed circuit will be below levels that are dangerous to humans.

[0049] FIG. **3** is a perspective view of one example of an electrically-powered vehicle **100** with which embodiments of the present disclosure may be used. In this example, vehicle **100** may be a passenger or transit bus, though as noted above embodiments of the present disclosure may be used with many other types of electrically-powered vehicles. It will be appreciated that vehicle **100** comprises a body **102** supported by a wheeled chassis in a body-on-frame construction. However, the present disclosure is not so limited, and in some embodiments vehicle **100** may have a unibody or monocoque chassis construction. As described in more detail below, vehicle **100** also comprises a propulsion system comprising at least one electric motor that is also supported by the wheeled chassis and may comprise individually dedicated motors for the respective wheels. The at least one electric motor drives wheels **104** of the wheeled chassis. Additionally, and also as described below, vehicle **100** comprises an electric power source supported by the wheeled chassis and in electrical communication with the propulsion system. In various embodiments, the electric power source comprises one or more battery packs. Examples of battery packs may be analogous to the lithium ion battery packs offered by various manufacturers, including Cummins Inc. of Columbus, Indiana. In one particular embodiment, vehicle **100** is a battery electric bus that is entirely electrically-powered. In other embodiments, vehicle **100** may be a hybrid power bus employing an engine-generator in communication with one or more suitable battery packs.

[0050] FIGS. **4A-4C** present a block diagram of one example of a high-voltage and accessory architecture **200** of an electrically-powered vehicle with which embodiments of the present disclosure may be used. In particular, architecture **200** comprises the high-voltage components and other electronic accessories of a battery-electric bus, and thus architecture may be used in the electrically-powered vehicle **100** of FIG. **3**. As shown, architecture **200** comprises an energy storage system **202**. Energy storage system **202** includes a plurality of battery packs, and in this illustrated embodiment six such battery packs **204-214** are provided. Battery packs **204-214** may be, e.g., BP74E Li-ion 74 kWh battery packs offered by Cummins Inc. Battery packs **204-214** may be disposed in a variety of locations on the electric bus and all possibilities are intended to be within the spirit and scope of the present disclosure. In one example, battery packs **204-208** may be disposed on a roof of the battery-electric bus, battery packs **210-212** may be disposed in a powertrain compartment of the bus, and battery pack **214** may be disposed on a lower forward location of the wheeled chassis. In one embodiment, each of battery packs **204-214** comprise two switches, e.g., electrical contactors, inside each battery pack, one for the positive high-voltage output and one for the negative high-voltage output. As will be appreciated, both such contactors must be closed to complete the circuit for high voltage to be available outside of the battery pack. [0051] In any event, battery packs **204-214** are all in electronic communication with an energy storage system junction box **216**. Certain aspects of the structure and electrical circuitry of a junction box analogous to junction box **216** in accordance with an embodiment of the present disclosure are discussed in greater detail below with reference to FIGS. **7-8**. In general, however, junction box **216** receives and electrically joins positive and negative electrical connections from respective terminals of each battery pack **204-214**. Also as described below, junction box **216** comprises two switches, e.g., electrical contactors, respectively in electrical communication with the positive and negative high voltage legs, that may serve to further isolate energy storage system **202** from the rest of the system, including the vehicle chassis. The remaining components of architecture **200** are in electrical communication with energy storage system **202** via direct or indirect connections to junction box **216**. In other words, the remaining components, which may be viewed as an electrical load, are electrically disposed in parallel with respect to junction box **216** and with respect to energy storage system **202**.

[0052] Those of skill in the art are familiar with high-voltage components and accessories in an electrical system of an electrically powered bus, and thus such components are not described in

detail herein. In general, though, architecture **200** comprises a battery thermal management system **218** and a coolant heater **220** in electrical communication with junction box **216**. Also, in this embodiment, architecture **200** comprises a high voltage direct current (HVDC) junction box **222**. As with junction box **216** above, junction box **222** may comprise two switches, e.g., electrical contactors, which serve to disconnect energy storage system **202** from downstream system components.

[0053] Certain components of architecture **200** are in electrical communication with junction box **216** via HVDC junction box **222**. For example, architecture **200** comprises a motor inverter **224** that is in electrical communication with HVDC junction box **222**. Motor inverter **224** provides alternating current to a traction motor **226**, which drives the wheels of the bus. Additionally, a plurality of charging inlets **228** are in electrical communication with HVDC junction box **222**. Further, HVDC junction box **222** is in electrical communication with an air compressor **230** and a heating, ventilation, and air conditioning (HVAC) system **232**. A DC/AC inverter **234** that is electrically disposed between air compressor **230** and HVAC system **232** and HVDC junction box **222** provides AC power to these components, e.g., via a high-voltage alternating current junction box **236**. Finally, architecture **200** comprises a low voltage direct current (LVDC) junction box **238** that receives low voltage DC power, e.g., via electrical communication with one or more DC/DC converters **240**. In this example, low-voltage DC power may comprise 24 VDC power. Various low-voltage components and/or loads may be in electrical communication with LVDC junction box **238**, such as an electro-hydraulic power (EHP) steering pump **242**, house battery charging **244**, house loads **246**, an ECP cooling system **248**, and an HVAC low voltage connection **250**.

[0054] Electrically-powered vehicles such as those described above with respect to FIGS. 3 and 4A-4C may typically include one or more switches that, when actuated to be placed in an open position, operate to open at least part of the vehicle's high-voltage electrical circuitry so that high-voltage power is not applied to at least part of the electrical load. As noted above, an operator may actuate such switch(es) as part of an LOTO or other safety procedure used to create an electrically safe work condition.

[0055] In this regard, an electrically-powered vehicle may comprise a high-voltage interlock loop (HVIL). Very generally, the HVIL may comprise a low voltage circuit that controls one or more switches (e.g., breakers, contactors, and disconnects) electrically disposed within the vehicle's high-voltage circuitry. FIGS. 5A-5C present a circuit diagram of one example of a HVIL **300** that may be used with the high-voltage and accessory architecture **200**. Other examples of HVILs may be used with the high-voltage and accessory architecture **200** and all such examples are intended to be within the spirit and scope of the present disclosure.

[0056] In the illustrated embodiment, HVIL **300** comprises a circuit that passes in series through certain high-voltage connections on an electrically-powered vehicle, such as the battery-electric buses described above. More particularly, and by way of example, HVIL **300** comprises a system control module (SCM) **302** that is electrically connected in series with a variety of high-voltage components, including but not limited to chargers **228**, traction motor **226**, motor inverter **224**, battery thermal management systems **218**, energy storage system junction box **216**, HVDC junction box **222**, DC/DC converters **240**, DC/AC inverter **234**, high-voltage AC junction box **236**, HVAC **232**, and low voltage junction box **238**. In particular, each high-voltage component may comprise a high-voltage connector that comprises an HVIL circuit. The HVIL circuit may be a "last mate first break" (LMFB) style circuit in that it may open as soon as the connector is about to be disconnected before the main conductor is disconnected. Although not shown in FIGS. 5A-5C, battery packs **202-214** may each comprise an internal HVIL circuit that is part of HVIL **300**. Further, HVIL **300** may be in electrical communication with switches on the interior of component lids and access door panels, though these too are not shown in FIGS. 5A-5C.

[0057] In one embodiment, SCM **302** outputs a low voltage signal onto the HVIL **300** circuit. SCM **302** monitors HVIL **300** circuit to determine whether the signal is received. Other devices may

monitor subsections of the circuit for fault analysis and troubleshooting assistance. Also, each battery pack **204-214** has HVIL detection within the battery pack and reports the HVIL status back to SCM **302**. If any component or switch on the HVIL **300** circuit is disconnected or opened, then the HVIL may be broken, and the SCM **302** may measure an open circuit. SCM **302** may then cause the high-voltage contactors within the high-voltage circuitry (e.g., the high-voltage contactors located in energy storage system junction box **238**, HVDC junction box **222**, and/or in each battery pack **204-214**) to change state (e.g., from a closed position to an open position), thereby isolating all high voltage to battery packs **204-214**. As a result, if an individual or operator attempts to access a high-voltage electrical component, e.g., by opening an access door or component lid, HVIL **300** should be opened, and SCM **302** should disconnect energy storage system **202** from the other high-voltage system circuitry and components.

[0058] Other switches may be present in the electrically-powered vehicle. For example, battery packs **204-214** of energy storage system **202** may each have a removable manual service disconnect (MSD), the removal of which disables electric current flow from a given battery pack. As shown in FIG. **6**, low-voltage junction box **238** may comprise a manual disconnect switch **304**. An operator may move switch **304** from the “ON” position to the “OFF” position as part of a LOTO procedure, and when switch **304** is in the “OFF” position, the operator may place a padlock **306** through a metal loop **308** extending from junction box **238** and an aperture defined in switch **304**. Thereby, switch **304** may remain “locked out” (e.g., unable to be moved to the “ON” position) until the operator returns with the key to padlock **306**. Actuation of switch **304** disconnects the low voltage circuitry of the electrically-powered vehicle. In one embodiment, because HVIL **300** may remain active as long as the low voltage system is connected, removing power to the low voltage circuitry may also cause HVIL **300** to open, thereby disconnecting the high-voltage components as described above.

[0059] In the past, while performing the LOTO procedure, the operator would need to take a number of safety precautions, including wearing protective safety gear and using proper measurement tools. Among other things, an operator may need high-voltage rated rubber gloves (e.g., 1000V rated gloves), leather gloves to be worn over the rubber gloves, safety glasses, and a suitably rated multimeter. Moreover, in the past, when the operator performed measurements to ensure that an electrically safe working condition has been created, the operator would need to wear a full-body arc-flash suit. With embodiments of the present disclosure, however, a full-body arc-flash suit may not be required.

[0060] In this regard, FIGS. **7** and **8** are perspective views of opposing sides of one example of an energy storage system junction box **310** comprising a test port **340** in accordance with an embodiment of the present disclosure. FIG. **9** is a detail view of junction box **310**. FIG. **10** is a perspective view of junction box **310**, wherein an operator is using a multimeter at test port **340** to determine the voltage drop across an electrical component of high-voltage circuitry to which junction box **310** is connected.

[0061] Energy storage system junction box **310** is configured for use in a high-voltage electrical system of an electrically-powered vehicle, such as high-voltage and accessory architecture **200**, described above. Accordingly, in the discussion that follows, reference numerals from components of architecture **200** are used to illustrate certain examples of electrical connections that may occur at junction box **310**. More particularly, junction box **310** comprises a box-shaped housing having four vertically-extending side walls **312-318** and a top wall **320** that together define an enclosure. Various physical connectors are provided on the housing of junction box **310**. For example, side wall **318** may support connectors **322** for the positive terminals of each battery pack **204-214**, and side wall **316** may support connectors **324** for the negative terminals of each battery pack **204-214**. On housing side wall **314**, connectors **326**, **328** are provided for the battery thermal management system **218** and the coolant heater **220**, respectively. Further, a connector **330** is provided on this side wall **314** that comprises the HVIL **300** into and out of junction box **310**, receives control

signals for the electrical contactors within junction box **310**, and receives low voltage power (24V) for junction box **310**. On side wall **312**, two sets of positive and negative output connectors **332**, **334** are provided to enable electrical communication between the positive and negative terminals of energy storage system **202** and HVDC junction box **222**. In particular, these four connectors are the high voltage outputs from the junction box **310** to the HVDC junction box **222**.

[0062] Additionally, side wall **312** may support a high voltage disconnect switch **336**. Switch **336** comprises, in this example, a rotary handle **338** that is lockable according to a LOTO protocol. When actuated, switch **336** may change state from a first (e.g., closed) position to a second (e.g., open) position. In the closed position, current may flow from energy storage system **202** across switch **336**, and in the open position, current may not flow from energy storage system **202**. Thus, actuation of switch **336** may disconnect the battery packs **204-214** from the downstream high-voltage system components, as indicated above with respect to switch **18** and FIG. **1**.

[0063] Junction box **310** comprises a test port **340** in accordance with an embodiment of the present disclosure. For example, test port **340** comprises a pair of electrical connectors **342**, **344** disposed on side wall **312**. In one embodiment, connectors **342**, **344** are IP rated connectors that each comprise a socket configured to receive a testing device (e.g., multimeter) test lead, and the connectors are disposed at the junction box by a distance of at least two inches. In some embodiments, a peripheral surface of connectors **342**, **344** may be threaded, and junction box **310** may comprise covers **346**, **348** that are threadably received over connectors **342**, **344**, respectively (See FIGS. **9-10**). Covers **346**, **348** may each be coupled with junction box **310** via chain or the like to prevent dropping or loss thereof. It will be appreciated that, in other embodiments, covers **346**, **348** may not be provided, and instead connectors **342**, **344** may be disposed behind a sealed access cover. Such a cover may, in some embodiments, be electrically connected with HVIL **300**.

[0064] In some embodiments, a body controller of the electrically-powered vehicle may be able to detect when one of covers **346**, **348** are removed from its respective connector **342**, **344**. For instance, covers **346**, **348** may be in electrical communication with the body controller via a circuit. When covers **346**, **348** are closed, the circuit is completed, and the body controller receives an input that indicates the covers **346**, **348** are coupled with connectors **342**, **344**. Thereby, for example, the body controller may disallow the bus from driving when covers **346**, **348** are not coupled with connectors **342**, **344**. This may prevent dust build-up, which can cause an isolation fault if it occurs. In one example, the covers **346**, **348** may not be electrically connected with HVIL **300** because if they were so connected, removal of covers **346**, **348** would cause high voltages to be isolated to the battery packs, and test port **340** could not be used (e.g., during a LOTO procedure as described herein) to confirm whether high voltages are or are not present.

[0065] In some example embodiments, circuitry within the energy storage junction box **310** may be analogous to that described with reference to FIG. **1** or FIG. **2**, above. Thus, for instance, the terminals of connectors **342** and **344** may be electrically connected to the high-voltage circuitry within junction box **310** in a manner similar to that described above with respect to terminals **20**, **22** in the embodiments of FIG. **1** or FIG. **2**.

[0066] Based on the foregoing, it will be appreciated that an operator may use test port **340** as part of a safety protocol to ensure that an electrically safe working condition exists in high-voltage circuitry to which junction box **310** is connected. For example, as part of an LOTO procedure, an operator may first remove covers **346**, **348** from connectors **342**, **344**, respectively. The operator may then insert the conductive tips of respective positive **388** and negative **390** test leads of a testing device, such as a multimeter **392** (FIG. **10**), into respective sockets such that an electrical connection is made between the test leads and respective terminals (e.g., terminals **20** and **22** in FIG. **1**). Doing so places the resistance of multimeter **392** electrically in parallel with both the electric power source and the electrical load (via connectors **332**, **334**), such that the voltage drop across the resistance of multimeter **392** may be directly related to the open circuit voltage of the electric power source (depending, of course, on the position/state of switches upstream from the

terminals, such as electronic disconnects, electrical contactors within battery packs **204-214**, and/or the MSDs in the battery packs **204-214**), and subject to the impact of electrical components, such as resistors, within the circuitry of junction box **310**.

[0067] In one embodiment of a safety protocol, an operator may use test leads **388**, **390** of multimeter **392** to measure the respective voltage drops between the terminals. If, as shown in FIG. **10**, the voltage reading is above a predetermined threshold, such as five volts (5V) or fifty volts (50V), depending on the test resistance and otherwise the voltage divider network, then the operator may determine that high voltage is present within the system. Notably, in performing these test measurements, in embodiments of the present disclosure the operator may only need to use a suitably rated multimeter, high-voltage rated gloves, and safety glasses when taking measurements. Again, because of the design of test port **340** and the configuration of the circuitry within junction box **310**, in the event of a short circuit between one of the terminals and ground or between one of the terminals and the electrical power source, the current flowing through such short circuit will be below levels that are dangerous to humans.

[0068] The operator may proceed to close various switches as part of the LOTO protocol, such as the low-voltage junction box **238** manual disconnect switch **304** and the high voltage disconnect switch **336** of junction box **310**. In some embodiments, the operator may be able to see the voltage dropping on the multimeter **392** display as the LOTO protocol is implemented. The operator may then (or after a predetermined amount of time) repeat the same test measurement across the terminals. If the voltage measured is below the predetermined threshold, then the operator may determine that high-voltage has been safely isolated to the battery packs **204-214** and removed from the high-voltage components downstream therefrom. Thus, the operator will have verified that an electrically safe working condition exists.

[0069] Those of skill in the art will appreciate that other benefits derive from embodiments of the present disclosure. For instance, in some embodiments, the electrical configuration of the test port may cause only a minimal voltage drop at the test instrument. Further, prior to the present disclosure, an operator was sometimes required to remove the MSDs from each of the battery packs of a vehicle's energy storage system as part of the safety protocol to ensure that an electrically safe working condition had been created. Doing so is generally undesirable, though. For example, gaining access to all battery packs of an electrically-powered vehicle can be cumbersome and time consuming. On an electrically-powered bus, for instance, one or more battery packs may be located in/on the roof of the bus, one or more battery packs may be located at the rear of the bus chassis, and one or more battery packs may be located in a forward position on the chassis. Moreover, removing the MSDs may cause a vehicle's control system to register fault codes and errors that must be cleared by the operator before the vehicle can be returned to normal operation. Further, the MSDs have a lifetime of approximately five hundred (500) insertion and removal cycles. In embodiments of the disclosure described above, though, the combination of the high voltage disconnect switch **336** of junction box **310** and test port **340** means that an operator can safely work on the high-voltage system downstream of junction box **310** without having to remove the MSDs from each of the battery packs **204-214**.

[0070] Additionally, in various embodiments, an isolation meter may be used with test port **340**/connectors **342** and **344** to perform an isolation check. As those of skill in the art will appreciate, additional isolation checking may be required if the test indicates a failure or an unclear result, or if other indicators from battery measurement points are indicating isolation failure separately. In one embodiment, test port **340**/connectors **342** and **344** may be used to measure the overall system resistance to determine if a leakage issue (e.g., isolation fault) is present. For instance, leakage may be occurring if there is a path between high voltage positive to ground, high voltage negative to ground, or high voltage positive to high voltage negative.

[0071] In various embodiments, a testing device, such as but not limited to a multimeter, may be built into or otherwise disposed in a junction box containing a test port or otherwise part of the

electric bus, in a position so that the testing device's output screen is visible to an operator. For instance, where the testing device is disposed within the junction box, a transparent window may be provided in the junction box so that an operator can view the testing device's output data from a position outside the junction box. Accordingly, the testing device may always be present or, if not permanent, may be disposed in the junction box for some period of time, and the operator will not need to be concerned with connecting the testing device. The operation of such an arrangement, from the standpoint of current flow in the event of short circuits, is similar to that described above with regard to FIG. 1, where the test instrument's resistance corresponds to resistance 44 in FIG. 1. [0072] Based on the foregoing, it will be appreciated that embodiments of the disclosure provide improved systems and methods for ensuring that a high-voltage electrical system is in an electrically safe working condition. Many modifications and other embodiments of the disclosure set forth herein will come to mind to one skilled in the art to which the disclosure pertains having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the disclosure is not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing descriptions and the associated drawings describe example embodiments in the context of certain example combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. In cases where advantages, benefits or solutions to problems are described herein, it should be appreciated that such advantages, benefits and/or solutions may be applicable to some example embodiments, but not necessarily all example embodiments. Thus, any advantages, benefits or solutions described herein should not be thought of as being critical, required or essential to all embodiments or to that which is claimed herein. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A high-voltage electrical system, comprising: a high-voltage electric power source; an electrical load disposed electrically across the electric power source so that the electric power source provides electric power to the electrical load; a first electrical terminal; a second electrical terminal; a first at least one electrical component electrically disposed between the first electrical terminal and the electric power source and between the first electrical terminal and the electrical load; a second at least one electrical component electrically disposed between the second electrical terminal and the electric power source and between the second electrical terminal and the electrical load; at least one switch disposed electrically between the electric power source and the electrical load and at least one of the first electrical terminal and the second electrical terminal so that the at least one switch, in a first switch state, conducts electric current between the electric power source and the electrical load and at least one of the first electrical terminal and the second electrical terminal and, in a second switch state, blocks flow of electric current from the electric power source to at least part of the electrical load; and a junction box disposed between the high-voltage electric power source and the electrical load and having a housing that encloses the first electrical terminal, the second electrical terminal, the first at least one electrical component, and the second at least one electrical component; wherein the first at least one electrical component is configured so that the first at least one electrical component passes at most a low level electric current to the first electrical terminal when a short circuit between the first electrical terminal and a portion of an electric circuit including the electric power source and the electrical load that has a high voltage

level completes an electric circuit that includes the first at least one electrical component and the electric power source, and wherein the second at least one electrical component is configured so that the second at least one electrical component passes at most a low level electric current to the second electrical terminal when a short circuit between a portion of an electric circuit including the electric power source and the electrical load that has a high voltage level completes an electric circuit that includes the second at least one electrical component and the electric power source.

2. The high-voltage electrical system as in claim 1, wherein the second at least one electrical component is electrically disposed across the electric power source with respect to the first at least one electrical component.
3. The high-voltage electrical system as in claim 2, wherein the electrical power source is a battery system and wherein the first electrical terminal is connected to a positive pole of the battery system and the second electrical terminal is connected to a negative pole of the battery system.
4. The high-voltage electrical system as in claim 2, wherein the first at least one electrical component comprises a resistor.
5. The high-voltage electrical system as in claim 2, wherein the second at least one electrical component comprises a resistor.
6. The high-voltage electrical system as in claim 2, wherein the first at least one electrical component and the second at least one electrical component comprise a transformer having a primary winding disposed electrically in parallel with the power source and a secondary winding electrically connecting the first electrical terminal and the second electrical terminal.
7. The high-voltage electrical system as in claim 1, wherein the high-voltage electric power source comprises one or more batteries together capable of storing at least 74 kWh of energy.
8. The high-voltage electrical system as in claim 7, wherein the high-voltage electric power source comprises one or more batteries together capable of storing at least 200 kWh of energy.
9. The high-voltage electrical system as in claim 1, further comprising a third at least one electrical component connected between a first position and a second position, wherein the first position is electrically between the first electrical terminal and the first at least one electrical component and wherein the second position is electrically between the second electrical terminal and the second at least one electrical component.
10. The high-voltage electrical system as in claim 9, wherein the third at least one electrical component is a resistor.
11. A bus, comprising: a body supported by a plurality of wheels; at least one electric motor disposed so that the at least one electric motor drives the plurality of wheels; a high-voltage electric power source in electrical communication with the at least one electric motor; an electrical load, including the at least one electric motor, disposed electrically across the electric power source so that the electric power source provides electric power to the electrical load; a first electrical terminal; a second electrical terminal; a first at least one electrical component electrically disposed between the first electrical terminal and the electric power source and between the first electrical terminal and the electrical load; a second at least one electrical component electrically disposed between the second electrical terminal and the electric power source and between the second electrical terminal and the electrical load; at least one switch disposed electrically between the electric power source and the electrical load and at least one of the first electrical terminal and the second electrical terminal so that the at least one switch, in a first switch state, conducts electric current between the electric power source and the electrical load and at least one of the first electrical terminal and the second electrical terminal and, in a second switch state, blocks flow of electric current from the electric power source to at least part of the electrical load; and a junction box disposed between the high-voltage electric power source and the electrical load and having a housing that encloses the first electrical terminal, the second electrical terminal, the first at least one electrical component, and the second at least one electrical component; wherein the first at least one electrical component is configured so that the first at least one electrical component

passes at most a low level electric current to the first electrical terminal when a short circuit between the first electrical terminal and a portion of an electric circuit including the electric power source and the electrical load that has a high voltage level completes an electric circuit that includes the first at least one electrical component and the electric power source, and wherein the second at least one electrical component is configured so that the second at least one electrical component passes at most a low level electric current to the second electrical terminal when a short circuit between a portion of an electric circuit including the electric power source and the electrical load that has a high voltage level completes an electric circuit that includes the second at least one electrical component and the electric power source.

12. The bus as in claim 11, wherein the second at least one electrical component is electrically disposed across the electric power source with respect to the first at least one electrical component.

13. The bus as in claim 11, wherein the first at least one electrical component and the second at least one electrical component each comprises one or more resistors, the one or more resistors having a resistance totaling at least 1 megaohm.

14. The bus as in claim 11, wherein the bus is a hybrid power bus comprising an engine-generator in communication with the high-voltage electric power source.

15. The bus as in claim 11, wherein the high-voltage electric power source comprises one or more battery packs and wherein the junction box comprises connectors for positive and negative terminals of each of the one or more battery packs.

16. The bus as in claim 9, wherein the junction box comprises connectors for one or more of a battery thermal management system, a coolant heater, and a high-voltage interlock loop.

17. The bus as in claim 11, wherein the junction box comprises a first connector in communication with the first electrical terminal and a second connector in communication with the second electrical terminal, wherein the first connector and the second connector each define a socket configured to receive a test lead of a multimeter.

18. The bus as in claim 11, further comprising a high voltage and accessory architecture comprising a plurality of high-voltage components and the electric power source, wherein the plurality of high-voltage components are in electrical communication with the electric power source via direct or indirect connection to the junction box.

19. A high-voltage electrical system, comprising: a high-voltage electric power source; an electrical load disposed electrically across the electric power source so that the electric power source provides electric power to the electrical load; a first electrical terminal and at least one first electrical component; a second electrical terminal and at least one second electrical component; at least one switch disposed electrically between the electric power source and the electrical load and at least one of the first electrical terminal and the second electrical terminal so that the at least one switch, in a first switch state, conducts electric current between the electric power source and the electrical load and at least one of the first electrical terminal and the second electrical terminal and, in a second switch state, blocks flow of electric current from the electric power source to at least part of the electrical load; and a junction box disposed between the high-voltage electric power source and the electrical load and having a housing that encloses the first electrical terminal, the second electrical terminal, the at least one first electrical component, and the at least one second electrical component; wherein the first at least one electrical component is disposed and configured so that the first at least one electrical component passes at most a low level electric current to the first electrical terminal when a short circuit between the first electrical terminal and a portion of an electric circuit including the electric power source and the electrical load that has a high voltage level completes an electric circuit that includes the first at least one electrical component and the electric power source, and wherein the second at least one electrical component is disposed and configured so that the second at least one electrical component passes at most a low level electric current to the second electrical terminal when a short circuit between a portion of an electric circuit including the electric power source and the electrical load that has a high voltage level completes

an electric circuit that includes the second at least one electrical component and the electric power source.

20. The high-voltage electrical system of claim 19, wherein application of a predetermined resistance across the first and second electrical terminals produces a voltage drop across the predetermined resistance that is in a voltage drop first state when the at least one switch is in its first switch state and is in a voltage drop second state when the at least one switch is in its second switch state, wherein the voltage drop first state is different from the voltage drop second state.
